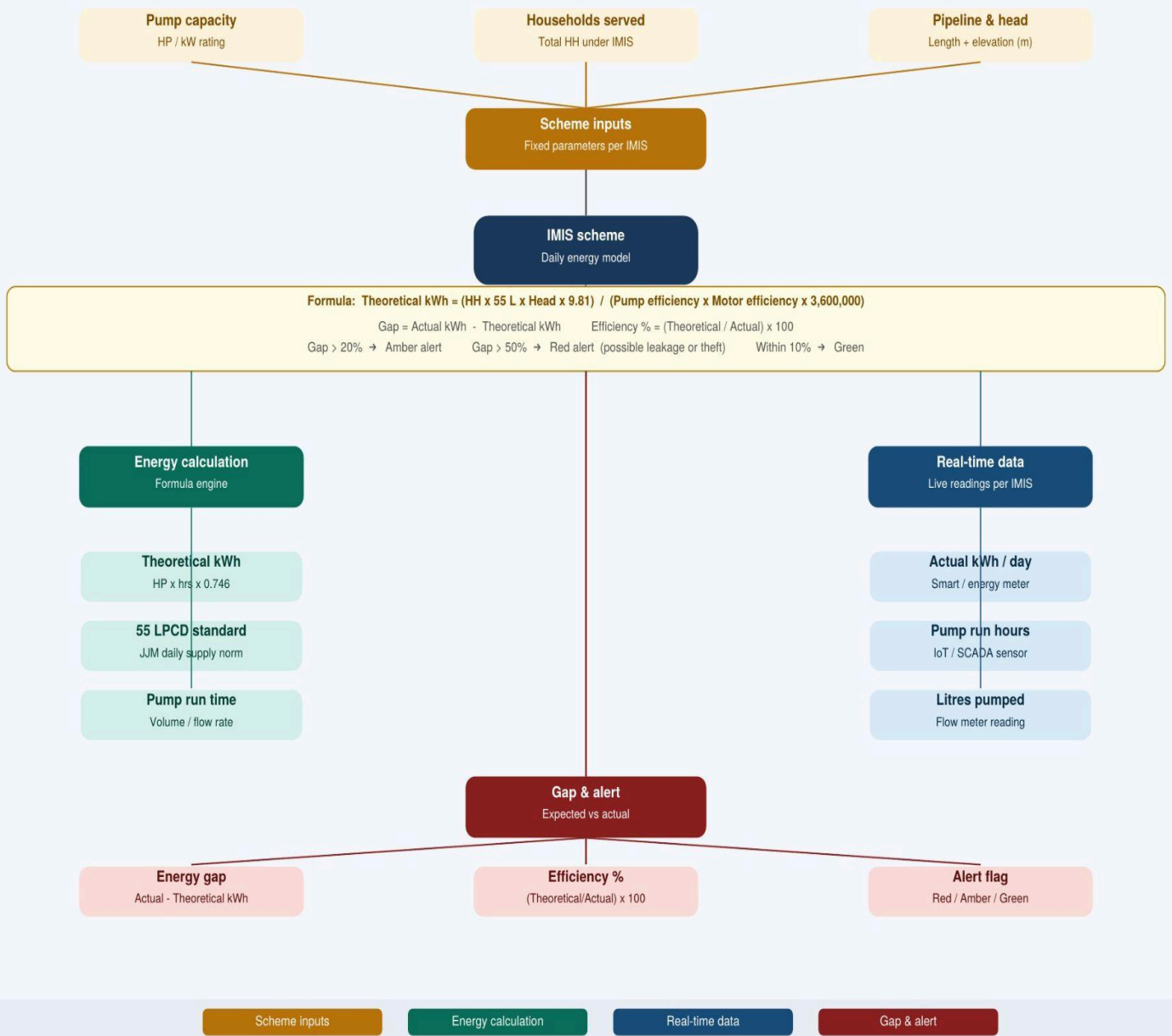


IMIS Scheme — Energy Consumption Monitoring Model

Theoretical daily kWh vs Real-time actual kWh | Per IMIS scheme



Scheme Inputs (top), the fixed facts about the scheme:

These are things that don't change day to day. They're the physical characteristics of the scheme pulled from JJM IMIS portal.

- **Pump capacity (HP/kW):** this is the engine of the scheme. A bigger pump consumes more electricity. Without knowing HP, you literally cannot calculate theoretical energy.
- **Households served:** this drives the *demand* side. JJM standard is 55 litres per person per day. More households = more water needed = more pump run time needed = more energy needed.

- **Pipeline & head:** head means the height the pump has to push water against (elevation + friction in the pipe). A scheme on a hill needs far more energy than a flat scheme with the same pump and same households. This is why two identical pumps in two different schemes can consume completely different amounts of energy.

Energy Calculation (left), the formula engine:

This is where the expected energy gets calculated using those fixed inputs.

- **Theoretical kWh:** the output of the formula. This is what the scheme *should* consume if it's running properly and supplying 55 LPCD to all households.
- **55 LPCD standard:** JJM benchmark. Every household is entitled to 55 litres per capita per day. This sets the minimum daily water volume the pump must deliver.
- **Pump run time:** calculated as: total volume needed ($HH \times 55L$) divided by the pump's flow rate. This tells you how many hours the pump needs to run.

Real-time Data (right), what actually happened today:

These come from live sensors on the ground.

- **Actual kWh/day:** straight from the energy meter. This is the ground truth of what was consumed.
- **Pump run hours:** from the IoT sensor. If the pump ran for 2 hours when it should have run for 4, either less water was supplied or the pump is oversized.
- **Litres pumped:** from the flow meter. This directly tells you if 55 LPCD was actually achieved or not.

Gap & Alert (bottom), the comparison engine:

This is where theoretical meets actuality.

- **Energy gap:** Actual kWh minus Theoretical kWh. If it's positive, more energy was used than expected. Could mean leakage, theft, pump degradation, or the pump running unnecessarily.
- **Efficiency %:** Theoretical divided by Actual $\times 100$. A score below 80% means the scheme is significantly wasteful. Above 90% means it's running well.
- **Alert flag:** the automatic decision based on the gap. Green within 10%, Amber at 20%+, Red at 50%+. Red usually means something is seriously wrong either physical leakage, energy theft, or equipment failure.

